

Laplace Transform

NOTATIONS: $r(t)$ general input signal
 $y(t)$ output signal
 $u(t)$ unit step function

For the function $f(t)$, the one-sided Laplace Transform is defined by:

$$F(s) = \mathcal{L}\{f(t)\} = \int_0^{\infty} f(t) e^{-st} dt \quad s \in \mathbb{C}$$

$$s = \sigma + j\omega \rightarrow e^{-st} = e^{-(\sigma + j\omega)t} = \underbrace{e^{-\sigma t}}_{\text{decay/growth}} \underbrace{e^{-j\omega t}}_{\text{oscillation}}$$

1. $\mathcal{L}$ constant function $f(t) = K$

$$\mathcal{L}\{K\} = \int_0^{\infty} K e^{-st} dt = \left[\frac{-K}{s} e^{-st} \right]_0^{\infty} = \frac{K}{s}$$

2. $\mathcal{L}$ exponential function $f(t) = e^{at}$

$$\mathcal{L}(e^{at}) = \int_0^{\infty} e^{at} e^{-st} dt = \int_0^{\infty} e^{-(s-a)t} dt = \frac{1}{s-a}$$

Example: $f(t) = 3e^{-2t}$

$$\mathcal{L}(f(t)) = \mathcal{L}\left\{3e^{-2t}\right\} = \int_0^{\infty} 3e^{-2t} e^{-st} dt =$$

$$= 3 \int_0^{\infty} e^{-(s+2)t} dt = \frac{3}{s+2}$$

3. What is the link to the complex numbers?

EULER $e^{j\omega t} = \cos(\omega t) + j\sin(\omega t)$

$$\mathcal{L}(e^{j\omega t}) = \frac{1}{s-j\omega} \text{ (from 2. above)}$$

$$\mathcal{L}(\cos(\omega t)) \stackrel{(*)}{=} \mathcal{L}\left\{\frac{e^{j\omega t} + e^{-j\omega t}}{2}\right\} \stackrel{(\ominus)}{=}$$

$$(*) \cos(\omega t) = \frac{e^{j\omega t} + e^{-j\omega t}}{2} = \frac{\cos(\omega t) + j\sin(\omega t) + \cos(-\omega t) + j\sin(-\omega t)}{2}$$

$$= \frac{2\cos(\omega t)}{2} \quad (\checkmark)$$

$$\stackrel{(\ominus)}{=} \frac{1}{2} \left(\frac{1}{s-j\omega} + \frac{1}{s+j\omega} \right) = \frac{1}{2} \left(\frac{(s+j\omega) + (s-j\omega)}{s^2 + \omega^2} \right) =$$

$$\mathcal{L}(\cos(\omega t)) = \frac{s}{s^2 + \omega^2}$$

$$\mathcal{L}(\sin(\omega t)) = \dots = \frac{\omega}{s^2 + \omega^2}$$

$\mathcal{L}\{k\} = \frac{k}{s}$	$\mathcal{L}\{e^{at}\} = \frac{1}{s-a}$	$\mathcal{L}\{\cos(\omega t)\} = \frac{s}{s^2 + \omega^2}$
		$\mathcal{L}\{\sin(\omega t)\} = \frac{\omega}{s^2 + \omega^2}$

4. Laplace power functions

$$\mathcal{L}\{t\} = \frac{1}{s^2}$$

$$\mathcal{L}\{t^n\} = \frac{n!}{s^{n+1}}$$
$$n! = n \cdot (n-1) \cdot (n-2) \cdot \dots \cdot 3 \cdot 2 \cdot 1$$

Example: $\mathcal{L}\{2 - 5e^{-3t}\} = \frac{2}{s} - \frac{5}{s+3}$

$$\mathcal{L}\{4 \sin 2t\} = 4 \cdot \frac{2}{s^2 + 4} = \frac{8}{s^2 + 4}$$

The Key properties: Linearity & Derivatives

$$\mathcal{L}\{af(t) + bg(t)\} = aF(s) + bG(s) \quad \text{LINEARITY}$$

$$\mathcal{L}\{e^{at} f(t)\} = F(s-a) \quad \text{EXPONENTIAL SHIFT}$$

The multiplication by e^{at} corresponds to a shift in the s domain. (DELAY)

$$F(s) = \mathcal{L}\{f(t)\} \rightarrow \mathcal{L}\{f'(t)\} = \int_0^{\infty} f'(t) e^{-st} dt$$

Integrate by parts:

$$\text{If: } u = e^{-st} \quad dv = f'(t) dt$$

$$\text{then: } du = -s e^{-st} dt \quad v = f(t)$$

$$\int_0^{\infty} f'(t) e^{-st} dt = \left[f(t) e^{-st} \right]_0^{\infty} + s \int_0^{\infty} f(t) e^{-st} dt = -f(0) + sF(s)$$

$$\mathcal{L}\{f'(s)\} = s F(s) - f(0)$$

DERIVATIVE

$$\mathcal{L}\{f''(s)\} = s^2 F(s) - s f(0) - f'(0)$$

$$\text{Assume: } f(0) = f'(0) = f''(0) = \dots = 0$$

$$\mathcal{L}\{f'(s)\} = s F(s)$$

$$\mathcal{L}\{f''(s)\} = s^2 F(s)$$

...

Example: FIRST-ORDER Initial Value Problem with $\mathcal{L}$

$$y' + 3y = 6 \quad y(0) = 1$$

$$\mathcal{L}\{y'\} + 3\mathcal{L}\{y\} = \mathcal{L}\{6\} \rightarrow sY(s) - 1 + 3Y(s) = \frac{6}{s}$$

$$\rightarrow Y(s)(s+3) = \frac{6}{s} + 1 \rightarrow Y(s) = \frac{s+6}{s(s+3)}$$

• PARTIAL FRACTION DECOMPOSITION

$$\frac{s+6}{s(s+3)} = \frac{A}{s} + \frac{B}{s+3} = \frac{A(s+3) + Bs}{s(s+3)}$$

$$s+6 = A(s+3) + Bs \quad \forall s$$

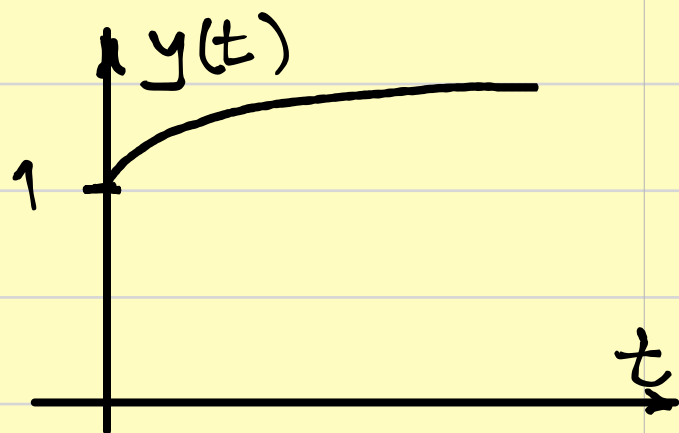
$$s=0 \rightarrow 0+6 = A(0+3) + B \cdot 0 \rightarrow A=2$$

$$s=-3 \rightarrow -3+6 = A(-3+3) + B(-3) \rightarrow B=-1$$

$$Y(s) = \frac{s+6}{s(s+3)} = \frac{2}{s} + \frac{(-1)}{s+3}$$

$$\mathcal{L}^{-1}\{Y(s)\} = y(t) = \mathcal{L}^{-1}\left\{\frac{2}{s}\right\} + \mathcal{L}^{-1}\left\{\frac{-1}{s+3}\right\}$$

$$y(t) = 2 - e^{-3t}$$



EXAMPLE: $y' + 2y = 4$ $y(0) = 0$

$$sY(s) - 0 + 2Y(s) = \frac{4}{s} \rightarrow Y(s)(s+2) = \frac{4}{s} \rightarrow$$

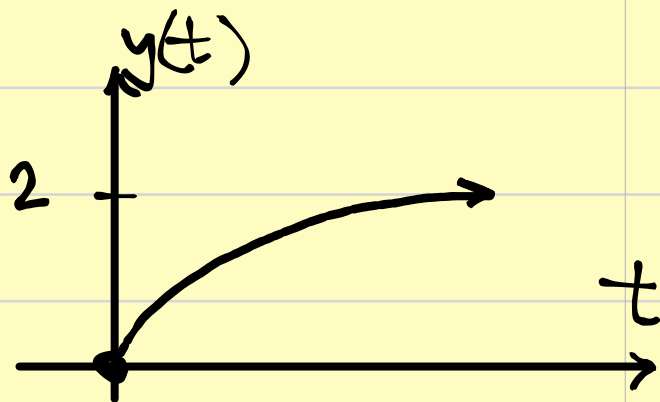
$$\rightarrow Y(s) = \frac{4}{s(s+2)} = \frac{A}{s} + \frac{B}{s+2} = \frac{2}{s} - \frac{2}{s+2}$$

$$4 = A(s+2) + Bs$$

$$s=0 \rightarrow 4 = 2A \rightarrow A=2$$

$$s=-2 \rightarrow 4 = -2B \rightarrow B=-2$$

$$\mathcal{L}^{-1}(Y(s)) = y(t) = 2 - 2e^{-2t}$$



EXAMPLE: Second-order initial value problem

$$y'' + 2y' + 5y = 0 \quad y(0) = 1 \quad y'(0) = 0$$

$$\mathcal{L}(y'') + 2\mathcal{L}(y') + 5\mathcal{L}(y) = 0$$

$$(s^2Y - sy(0) - y'(0)) + 2(sY - y(0)) + 5Y = 0$$

$$s^2Y - s + 2sY - 2 + 5Y = 0$$

$$(s^2 + 2s + 5)Y - (s + 2) = 0$$

$$Y(s) = \frac{s+2}{s^2+2s+5} = \frac{s+2}{(s-(-1+j2))(s-(-1-j2))}$$

$$s^2 + 2s + 5 = 0 \rightarrow \frac{-2 \pm \sqrt{4-20}}{2} = \frac{-2 \pm j.4}{2} = -1 \pm j2$$

$$Y(s) = \frac{A}{(s-(-1+j2))} + \frac{B}{(s-(-1-j2))}$$

$$s+2 = A(s-(-1-j2)) + B(s-(-1+j2))$$

$$s = -1-j2 : -1-j2+2 = A \cdot 0 + B(-1-j2-(-1+j2))$$

$$1-j2 = B \cdot (-j4) \rightarrow B = \frac{1-j2}{-j4} \cdot \frac{j.4}{j.4}$$

$$B = \frac{j.4+8}{16} \rightarrow B = \frac{2+j}{4}$$

$$s = -1+j2 : -1+j2+2 = A(-1+j2-(-1-j2)) + B \cdot 0$$

$$1+j2 = A \cdot 4j \rightarrow A = \frac{1+j2}{j4} \cdot \frac{-j4}{-j4}$$

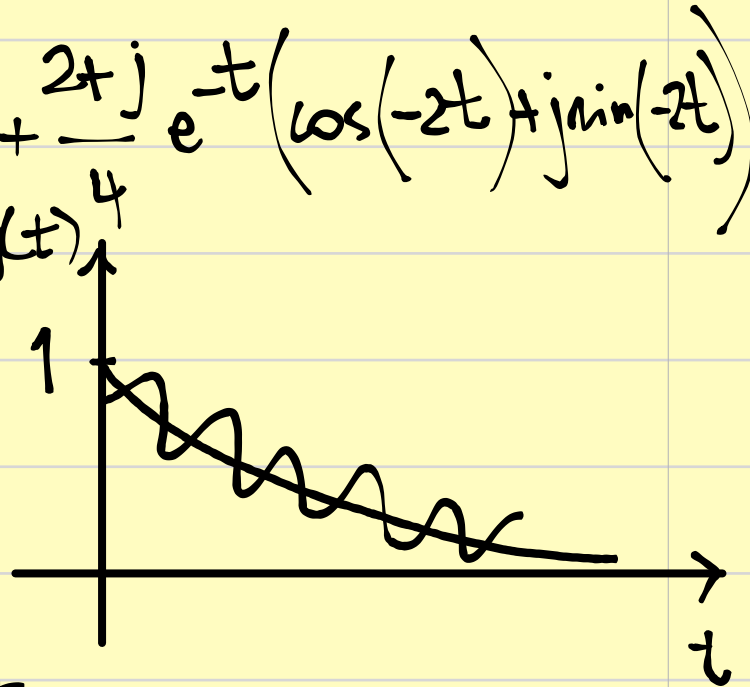
$$A = \frac{-j4+8}{16} \rightarrow A = \frac{2-j}{4}$$

$$Y(s) = \frac{2-j}{4} \cdot \frac{1}{s-(-1+j2)} + \frac{2+j}{4} \cdot \frac{1}{s-(-1-j2)}$$

$$\mathcal{L}^{-1}\{Y(s)\} = y(t) = \frac{2-j}{4} e^{(-1+j2)t} + \frac{2+j}{4} e^{(-1-j2)t}$$

$$= \frac{2-j}{4} \cdot e^{-t} \cdot (\cos 2t + j \sin 2t) + \frac{2+j}{4} e^{-t} (\cos(-2t) + j \sin(-2t))$$

$$= e^{-t} \cos 2t + \frac{1}{2} e^{-t} \sin 2t$$



Laplace Transform .. Dictionary

$$\frac{1}{s+a} \xrightleftharpoons[\mathcal{L}]{\mathcal{L}^{-1}} e^{-at}$$

$$\frac{1}{(s+a)^2} \xrightleftharpoons[\mathcal{L}]{\mathcal{L}^{-1}} t e^{-at}$$

$$\frac{s+a}{(s+a)^2 + \omega^2} \xrightleftharpoons[\mathcal{L}]{\mathcal{L}^{-1}} e^{-at} \cos \omega t$$

$$\frac{\omega}{(s+a)^2 + \omega^2} \xrightleftharpoons[\mathcal{L}]{\mathcal{L}^{-1}} e^{-at} \sin \omega t$$

